

Legacy Workflows

Local Tools & Data



Standalone Processes



Diverse Data Needs
& Repositories



Distinct Methodologies
& Assumptions



Future Enhancements



Cloud-Native Workflows



Modular Frameworks



Shared Data Repositories



Methodology Alignment
and Improvement

Risk Assessment Maturity Roadmap - DRAFT

July 16, 2026

Risk Assessment Maturity Roadmap

Contents

Risk Assessment Maturity Roadmap.....	i
Summary	1
QuickStart Guide	3
Risk Assessment Workstreams.....	5
Computation Methods	8
Uncertainty.....	12
Building & Infrastructure Inventory	15
Coastal Consequences	19
Inland Consequences	22
Downstream Uses.....	24
Recommended Immediate Start Activities.....	28

Figures

Figure 1-Bridging Legacy Investments & the Next-Generation Consequences Solution	2
Figure 2 - Process Flowchart Legend	3
Figure 3 - Inland vs. Coastal Flood Risk Assessment Methods.....	6
Figure 4 - Risk Assessment Methodology Roadmap	7
Figure 5 - Computation Methods Process Flowchart	10
Figure 6 - Legacy Workflows & Future Enhancements	11
Figure 7 - Flexible, Cloud-Native Architecture	11
Figure 8 - Uncertainty Process Flowchart	13
Figure 9 - Deterministic vs. Probabilistic Approaches	14
Figure 10 - Building & Infrastructure Inventory Process Flowchart.....	17
Figure 11 - Building and Infrastructure Inventory	18
Figure 12 - Coastal Consequences Process Flowchart	20
Figure 13 - Compound Flooding	21
Figure 14 - Impacts to Inland Consequences Methodology	23
Figure 15 - Downstream Uses Process Flowchart	26
Figure 16 - BCA Toolkit Integration.....	27

Tables

Table 1 - Stakeholder Descriptions	4
Table 2 - SOP Needs Descriptions	4
Table 3 - Computation Methods: Completed Milestones	9
Table 4 - Uncertainty: Completed Milestones.....	12
Table 5 – Building and Infrastructure Inventory: Completed Milestones	16
Table 6 - Coastal Consequences: Completed Milestones.....	19
Table 7 - Inland Consequences: Completed Milestones	22
Table 8 - Downstream Uses: Completed Milestones	25

List of Acronyms

AAL	Average Annual Loss
AI	Artificial Intelligence
API	Application Programming Interface
ArcGIS Pro	Esri ArcGIS Professional desktop GIS software
BCA	Benefit-Cost Analysis
BCAR	Benefit-Cost Analysis Re-engineering
CPFRA	Coastal Probabilistic Flood Risk Assessment
DDF	Depth-Damage Function
FEMA	Federal Emergency Management Agency
FFH	First Floor Height
FFRD	Future of Flood Risk Data
FHC	Flood Hazard Curve
FIRM	Flood Insurance Rate Map
HEC-FIA	Hydrologic Engineering Center – Flood Impact Analysis
HIFLD	Homeland Infrastructure Foundation-Level Data
IBC	International Building Code
NCIP	National Continuous Improvement Program
NFIP	National Flood Insurance Program

NFMP	National Flood Mapping Program
NHRAP	National Hazard Risk Assessment Program
NSI	National Structure Inventory
OpenHazard	Open-source implementation of Hazus methodologies
PFRA	Probabilistic Flood Risk Assessment
PTS	Production and Technical Services
QA	Quality Assurance
ROI	Return on Investment
SLTT	State, Local, Tribal, and Territorial
SOP	Standard Operating Procedure
SWL	Still Water Level
USACE	United States Army Corps of Engineers

Disclaimers

Please note that all proposed tasks are dependent upon funding and alignment on priorities from federal stakeholders.

Select graphics in this document were generated using artificial intelligence (AI) and were reviewed and edited by the authors for accuracy, clarity, and consistency with the report content.

Summary

This document was prepared for federal stakeholders to document the current state of FEMA's flood risk assessment processes and recommend updates, enhancements, and additional capabilities that support the transition to the Future of Flood Risk Data (FFRD) in support of FEMA's transition to the National Flood Mapping Program. The proposed tasks are intended to:

- Preserve proven methodologies.
- Streamline and modernize workflows in preparation for FFRD.
- Enhance, align, and standardize datasets, libraries, and supporting frameworks.
- Identify future enhancements that expand analytical capabilities.

Collectively, these efforts build on existing investments to bridge legacy capabilities with FEMA's next-generation flood consequence modeling solution (see **Figure 1**). Grounded in established FEMA and USACE scientific methodologies and existing operational workflows, the future solution is envisioned as secure, extensible, modular, enterprise-ready, and cloud-native, with a standardized Python implementation.

This report is organized around six primary workstreams:

- Computation Methods
- Uncertainty
- Building and Infrastructure Inventory
- Coastal Consequences
- Inland Consequences
- Downstream Uses

Although no new tasks are proposed specifically for the Inland Consequences workstream since that was a major emphasis during the [past year](#), it will be significantly influenced by advancements made across the other workstreams, particularly those related to computation methods, uncertainty, data models, and inventory management.

The report concludes with recommendations for activities that can begin immediately to support near-term progress while laying the foundation for longer-term FFRD implementation. Please note that all proposed tasks are subject to funding availability and prioritization by federal stakeholders.

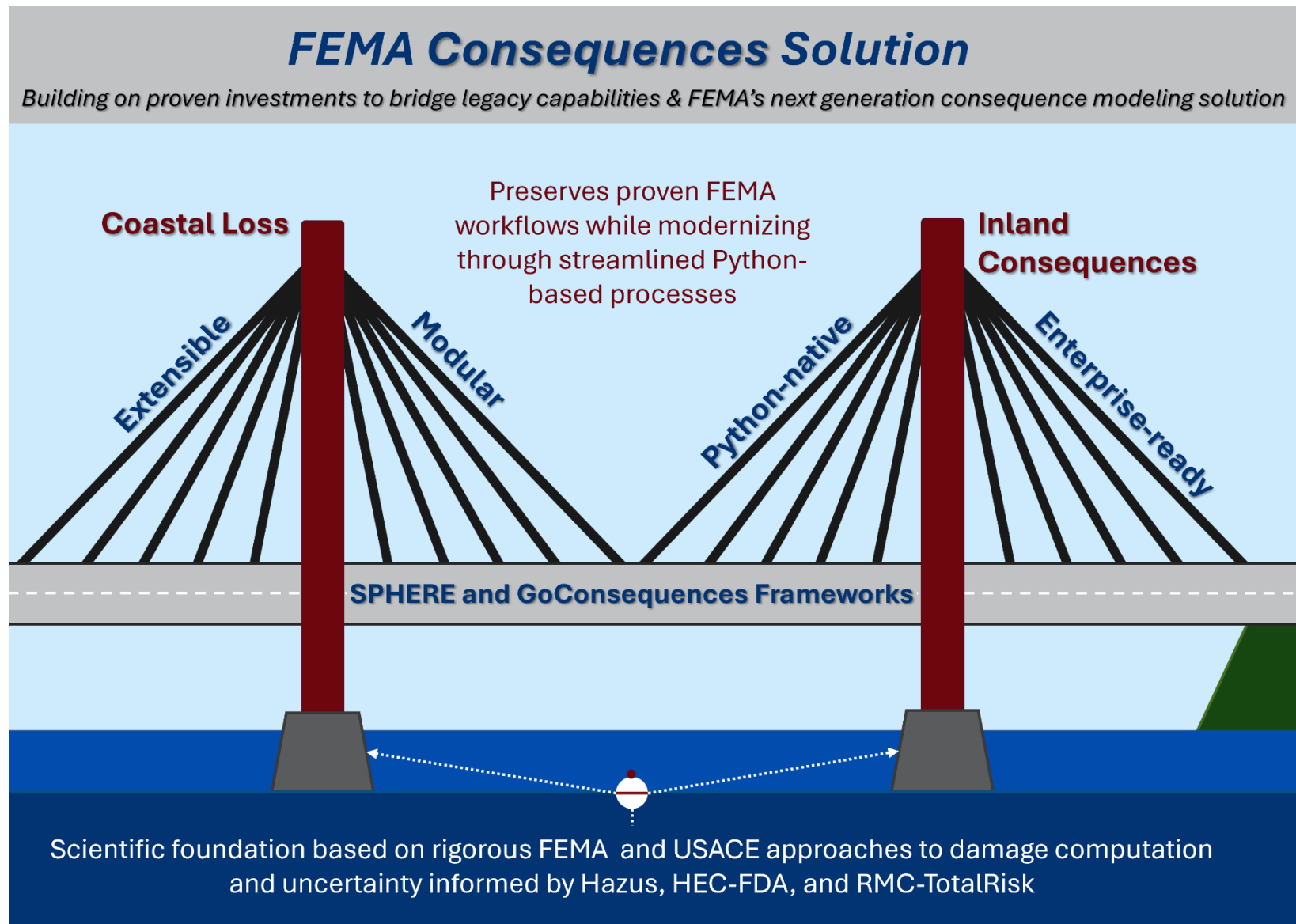


Figure 1 - Bridging Legacy Investments & the Next-Generation Consequences Solution

QuickStart Guide

Process Flowchart Legend

A proposed process flow has been developed for each workstream except Inland Consequences. Each flowchart includes a brief description of each task and is color-coded to indicate the proposed implementation timeframe (**Figure 2**). Associated stakeholders (**Table 1**) and required standard operating procedure (SOP) updates (**Table 2**) are also identified for each task.

Beneath each task, high-level objectives and key dependencies are provided to summarize the intended outcomes and relationships between activities. Although the tasks are presented in a linear sequence for readability, many can be executed concurrently where appropriate.



Figure 2 - Process Flowchart Legend

Stakeholders

Abbreviation	Description
BCA	Benefit-Cost Analysis team(s) and stakeholders
Cloud	FFRD Cloud Computing team(s)
Coastal	FEMA Coastal team(s)
Consequences	Consequence Solution team(s) and stakeholders
NFIP Actuaries	NFIP Actuarial stakeholders
NHRAP	National Hazard Risk Assessment Program team
USACE	United States Army Corps of Engineers stakeholders

Table 1 - Stakeholder Descriptions

SOP Needs

Abbreviation	Description
Baseline Convergence Criteria	Determine appropriate Monte Carlo convergence criteria references to include in the SOP
Coastal Method Update	Update the coastal loss methods to be followed in the SOP
Data Model Update	Update references to the proper data model or schema to be followed in the SOP
DDF Alignment	Update references to the DDF libraries to be used in the SOP
Default Assumptions	Provide default assumptions in the SOP
Describe Methods	Specify the methods to be followed in the SOP
Mapping Scheme Update	Update references to prescribed mapping schemes to be used in the SOP
Tiered Screening Guidance	Specify the screening criteria to be used to determine areas of additional compound flooding detail needs in the SOP

Table 2 - SOP Needs Descriptions

Risk Assessment Workstreams

Figure 3 compares the current state of the Inland and Coastal consequences tools, where they align and where they differ. While hazard data format and ingestion are different approaches necessitated by ADCIRC input, the analysis steps including damage functions are identical. Cross-walking Inland foundation types to align with deep and shallow foundation type categories are needed (e.g. Pile = Deep). **Figure 4** provides a high-level summary to the Risk Assessment Methodology Roadmap that highlights what has been completed and prioritized to be deployed.

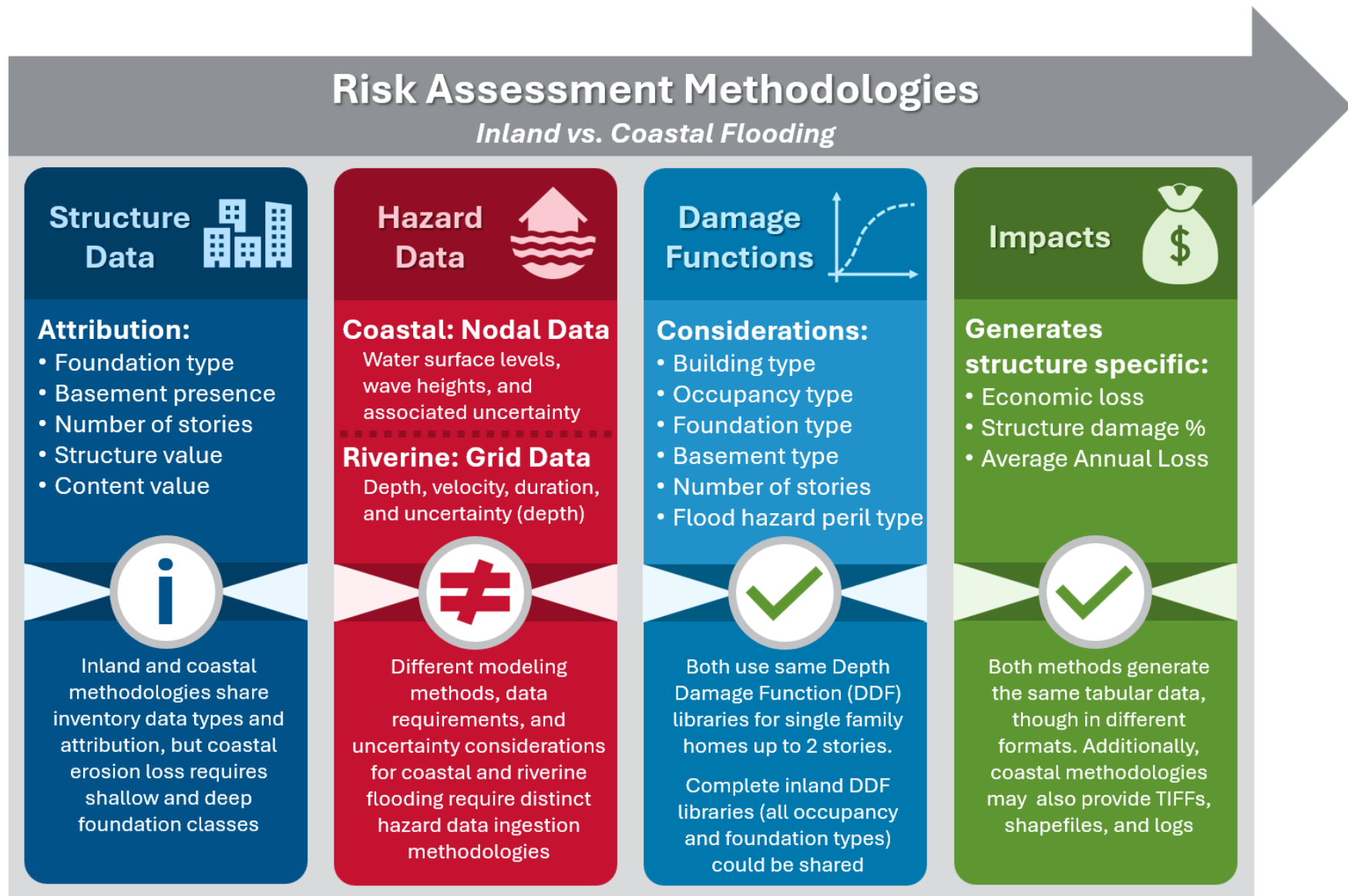


Figure 3 - Inland vs. Coastal Flood Risk Assessment Methods

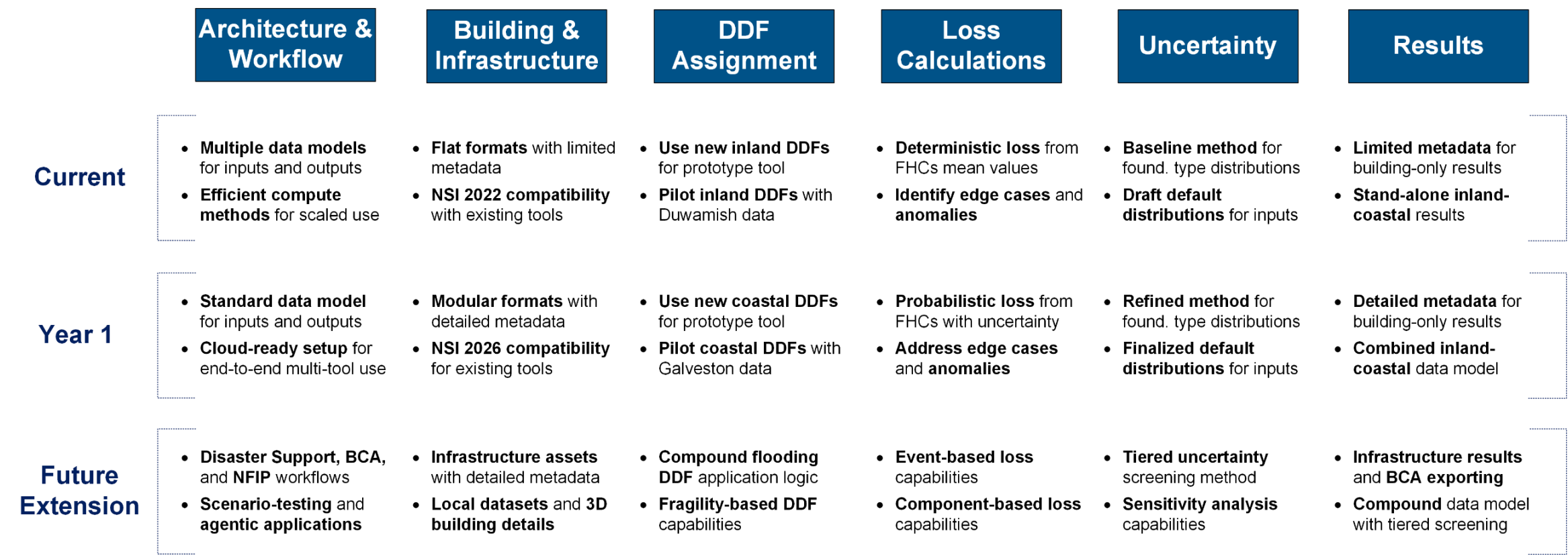


Figure 4 - Risk Assessment Methodology Roadmap

Computation Methods

Work To-Date

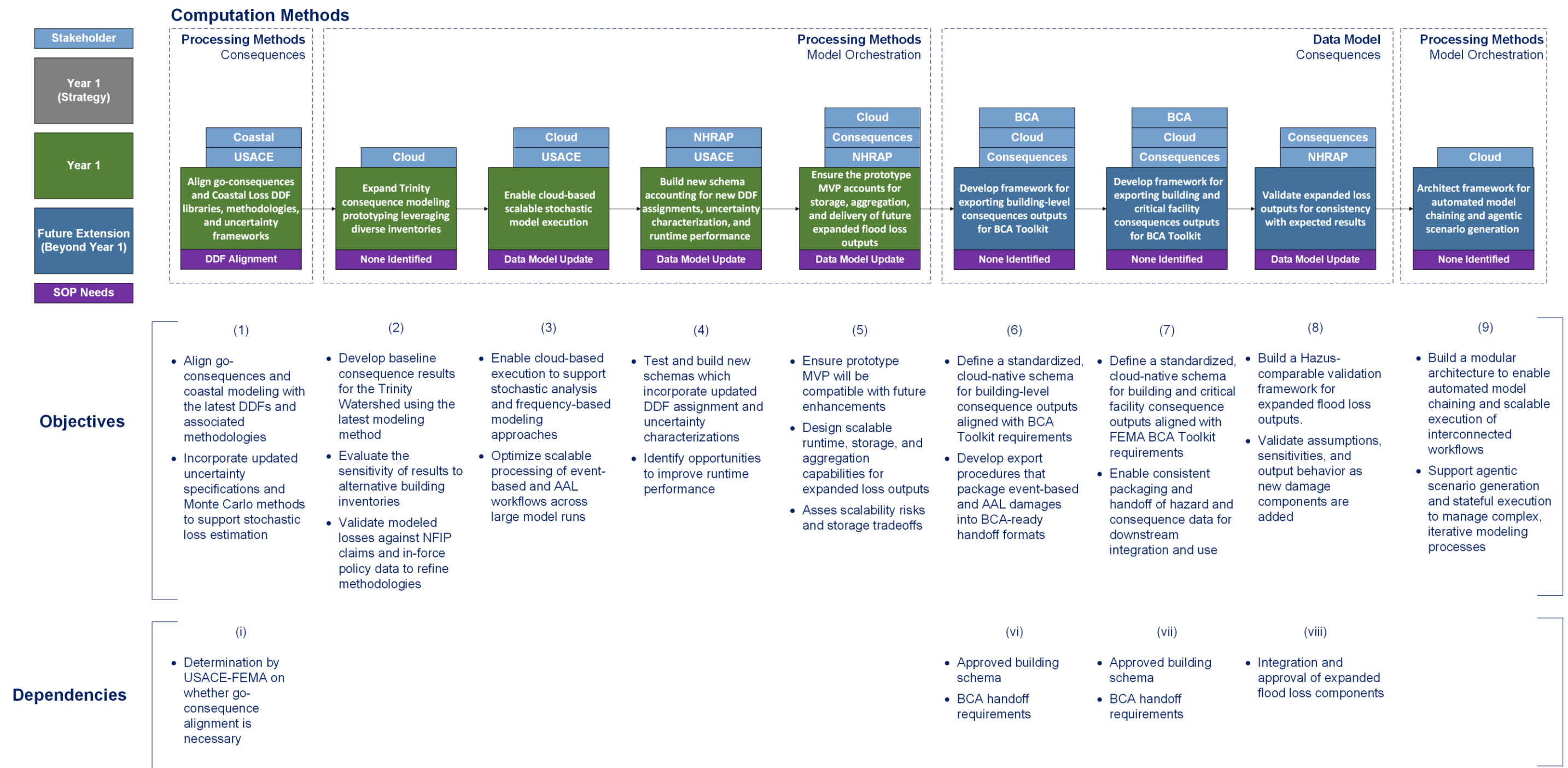
Damage depth function (DDF) methodologies have undergone substantial refinement over the past decade (**Table 3**), building on foundational FEMA guidance while expanding the breadth, consistency, and validation of available functions. Initial efforts incorporated FEMA Benefit-Cost Analysis Re-engineering (BCAR) updates to residential coastal DDFs and guidance, followed by the development of new single-family, multi-family, and inland residential DDFs to support Coastal and Inland Probabilistic Flood Risk Assessments. Subsequent work focused on evaluating uncertainty methods, developing a comprehensive DDF library, and establishing a more defensible damage function assignment methodology using flood-specific building classifications and revised foundation types. Recent efforts have emphasized validation and testing of these methodologies through comparisons with NFIP claims data, Hurricane Ian observations, and OpenHazard DDF libraries, resulting in recommendations to improve internal consistency, support updated riverine damage functions, and develop a single DDF library for operational implementation.

Proposed Work

Proposed tasks modernize consequence modeling workflows by transitioning from standalone processes with disparate data needs and libraries to a cloud-native, modular framework supported by a shared data repository and standardized methodologies (see **Figure 5**, **Figure 6**, and **Figure 7**). This work includes developing a new consequence schema that incorporates updated DDF assignments, improved uncertainty characterization, runtime and operational performance considerations, and flexibility for future enhancements. The framework also enables cloud-based execution to support stochastic analyses and frequency-based modeling approaches. Building on the work completed in the Duwamish pilot, the Trinity Watershed is proposed as a second testbed to establish a baseline using the latest consequence modeling methods. Subsequent efforts would leverage these baseline results to validate model performance, evaluate alternative building inventories, and compare modeled losses against NFIP claims and in-force policy data to further assess and refine the methodologies.

Table 3 - Computation Methods: Completed Milestones

Milestone	Year	Documentation
Initial Hazus DDF Library Development: Initial Hazus damage function library development and testing incorporating available FIA, USACE and FEMA DDFs	Pre- 2010	Hazus Archived Documentation
BCAR Updates: FEMA BCAR updates to Residential DDFs and Guidance for Coastal Flooding incorporated	2017	BCAR Flood Methodology-Updated Coastal DDFs
Coastal DDF Development: Development of Single-Family Residential DDFs for Coastal Probabilistic Flood Risk Assessment (CPFRA)	2020	Coastal DDFs Report
New and Expanded DDFs: Development of Multi-Family Low-Rise and Mid-Rise Residential DDFs for CPFRA. Expanded Single-Family Residential DDFs for Inland Probabilistic Flood Risk Assessment (PFRA)	2021	Coastal Multi-Family DDFs 2021 Inland Single Family Residential DDF Report
DDF Uncertainty Methods Review: Review of existing approaches and recommendations	2023	Uncertainty White Paper
DDF Library & Damage Function Assignment: Initial development of comprehensive DDF Library and propose new damage function assignment using flood-specific building types and revised foundation types	2024	OpenHazus Supporting Research Report – Task 1 Sec. 3
Validation Against Flood Insurance Claims Data: Exploration of Damage Functions with strong findings supporting new PFRA riverine functions and mixed results related to coastal claims data issues	2024	Flood Insurance Claims Data Exploration for Damage Functions
Validation of OpenHazus DDF library: Recommendations provided with a focus on internal consistency. DDF validation performed, leveraging detailed Hurricane Ian observations and claims data	2025	DDF Validation Report
Testing of OpenHazus DDF library: Performed using Duwamish FFRD data and finalization of library based on findings	2026	Inland Consequence Tool Report - Task 5 Sec. 8



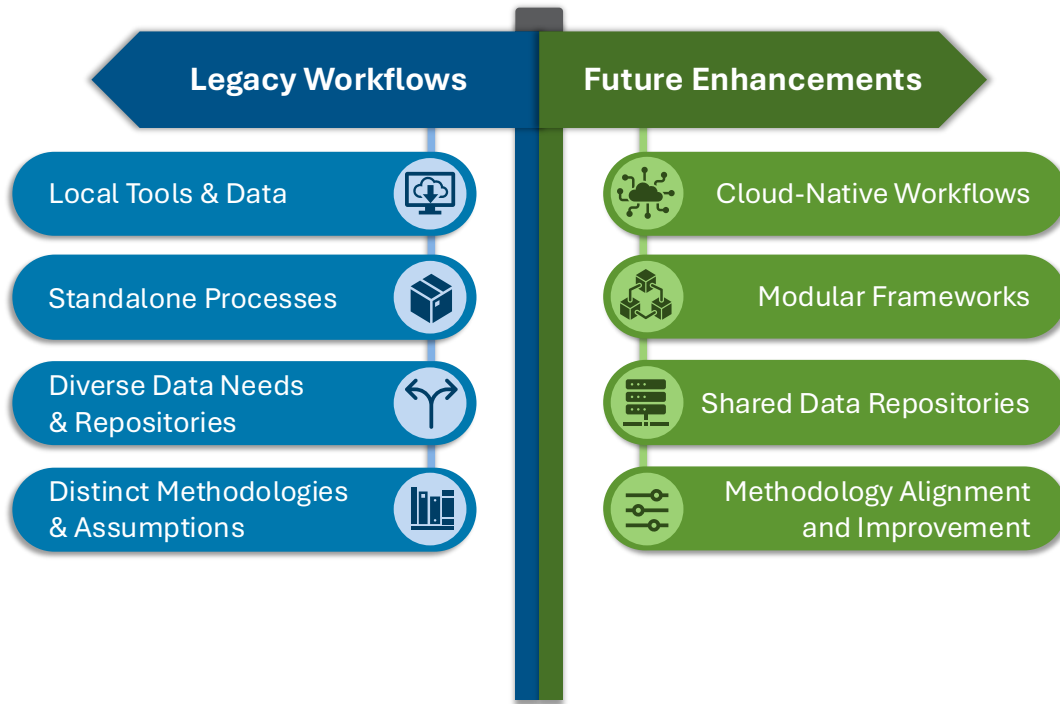


Figure 6 - Legacy Workflows & Future Enhancements

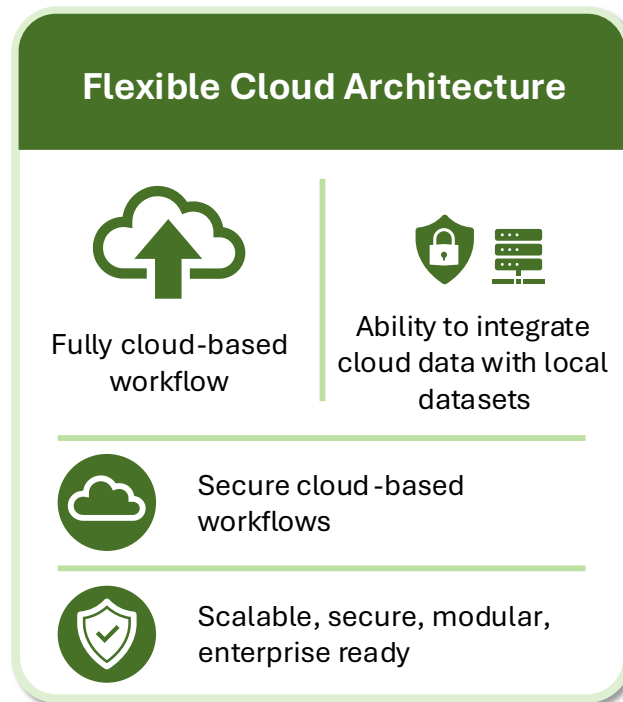


Figure 7 - Flexible, Cloud-Native Architecture

Uncertainty

Work To-Date

Prior efforts have investigated academic and industry best practices to reliably quantify uncertainty in the consequence components of the FFRD workflow (**Table 4**). This work has been focused on identifying key areas of uncertainty, methods to quantify them, and testing the importance of their influence on the loss outputs. Analysis was completed with research-grade code and resulted in proposed recommendations for further finalizing operational standards and assumptions for FFRD and downstream application use.

Proposed Work

Additional data model and method testing work is proposed to finalize the recommended default specifications when using these methods and facilitate efficient deployment within FFRD pipeline (**Figure 8** and **Figure 9**). Uncertainty methods should be flexible to easily switch between deterministic and more probabilistic methods per SOP or user preference within any loss modeling framework. Resulting schema standards and default assumptions would allow for integration into the FFRD SOP and consistent incorporation into NHRAP prototype tools and applications.

Table 4 - Uncertainty: Completed Milestones

Milestone	Year	Documentation
Hazard Uncertainty Methods: Discuss hazard data availability from FFRD and methods of constructing FHCs	2024	OpenHazard Supporting Research - Task 1
Computational Uncertainty Methods: Review existing tools and how parameterization is structured	2024	OpenHazard Supporting Research - Task 2
Built Environment Methods: Investigate deriving foundation type and FFH from real-world datasets	2024	OpenHazard Supporting Research - Task 3
Coastal Uncertainty Methods: Develop wave- and SWL-based computation for probabilistic loss	2025	Coastal FFRD AAL Calculation Tool
Mitigation Uncertainty Methods: Compare varying discount rate and distributional weighting assumptions	2025	OpenHazard Supporting Research - Task 4
DDF & Spatial Uncertainty Analysis: Test DDF and footprint methods with pilot in the Duwamish	2025	OpenHazard Supporting Research - Task 1
Built Environment Uncertainty Analysis: Test foundation type and FFH methods with pilot in the Duwamish	2025	OpenHazard Supporting Research - Task 2
Global Sensitivity Analysis: Test impact of interaction effects on overall loss with pilot in the Duwamish	2025	OpenHazard Supporting Research - Task 3
Probabilistic Output Needs: Identify how stakeholder needs and remaining gaps can inform next steps	2026	OpenHazard Supporting Research - Task 4

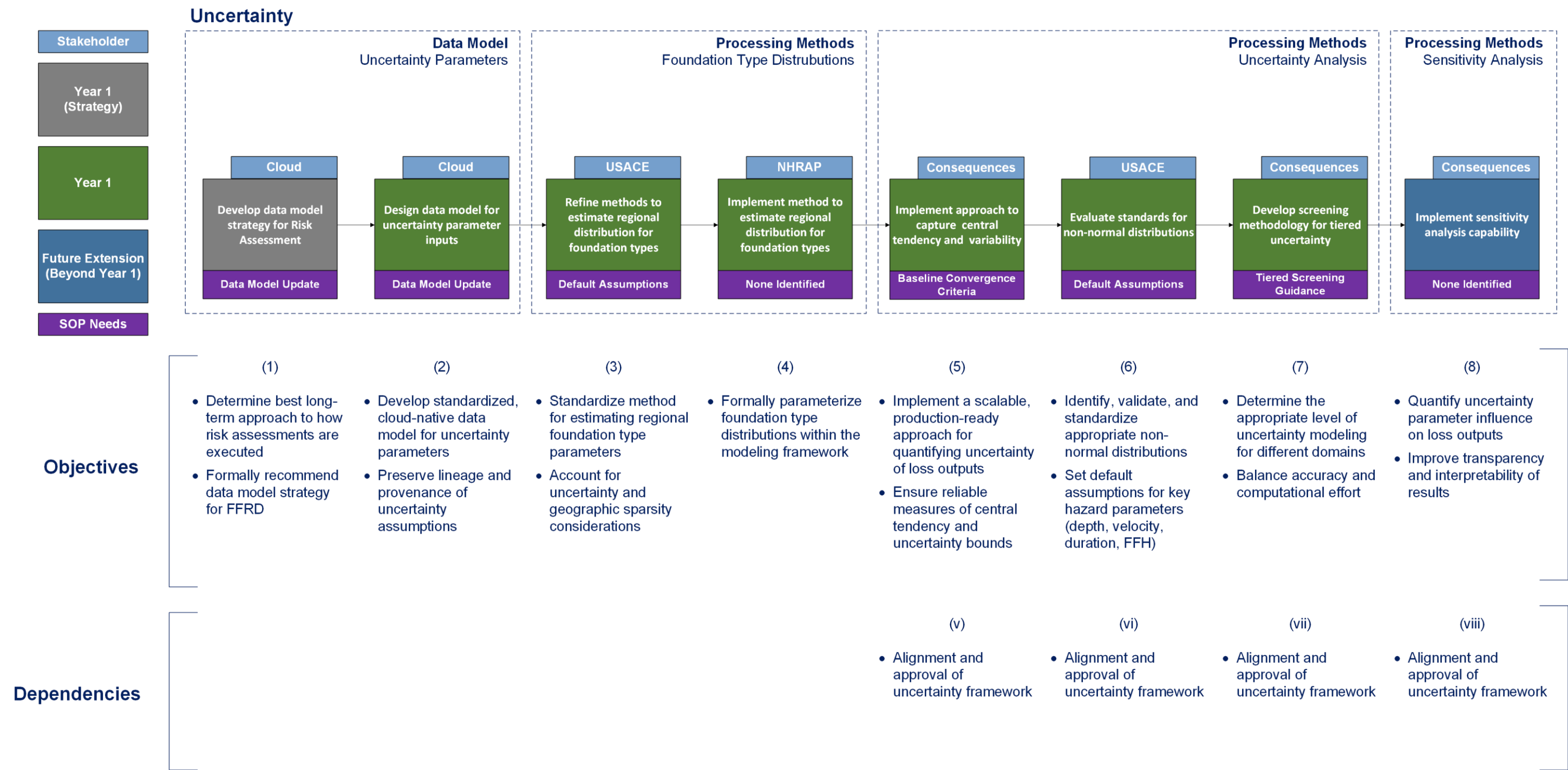


Figure 8 - Uncertainty Process Flowchart

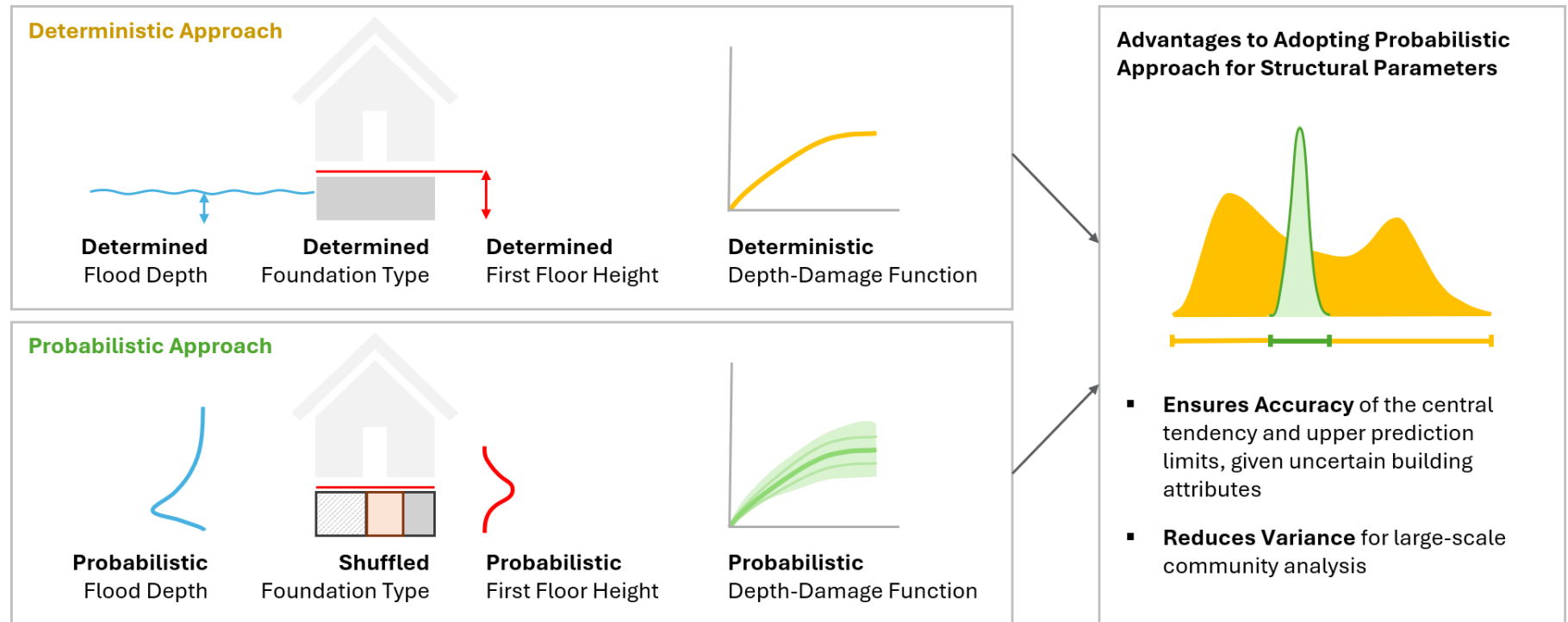


Figure 9 - Deterministic vs. Probabilistic Approaches

Building & Infrastructure Inventory

Work To-Date

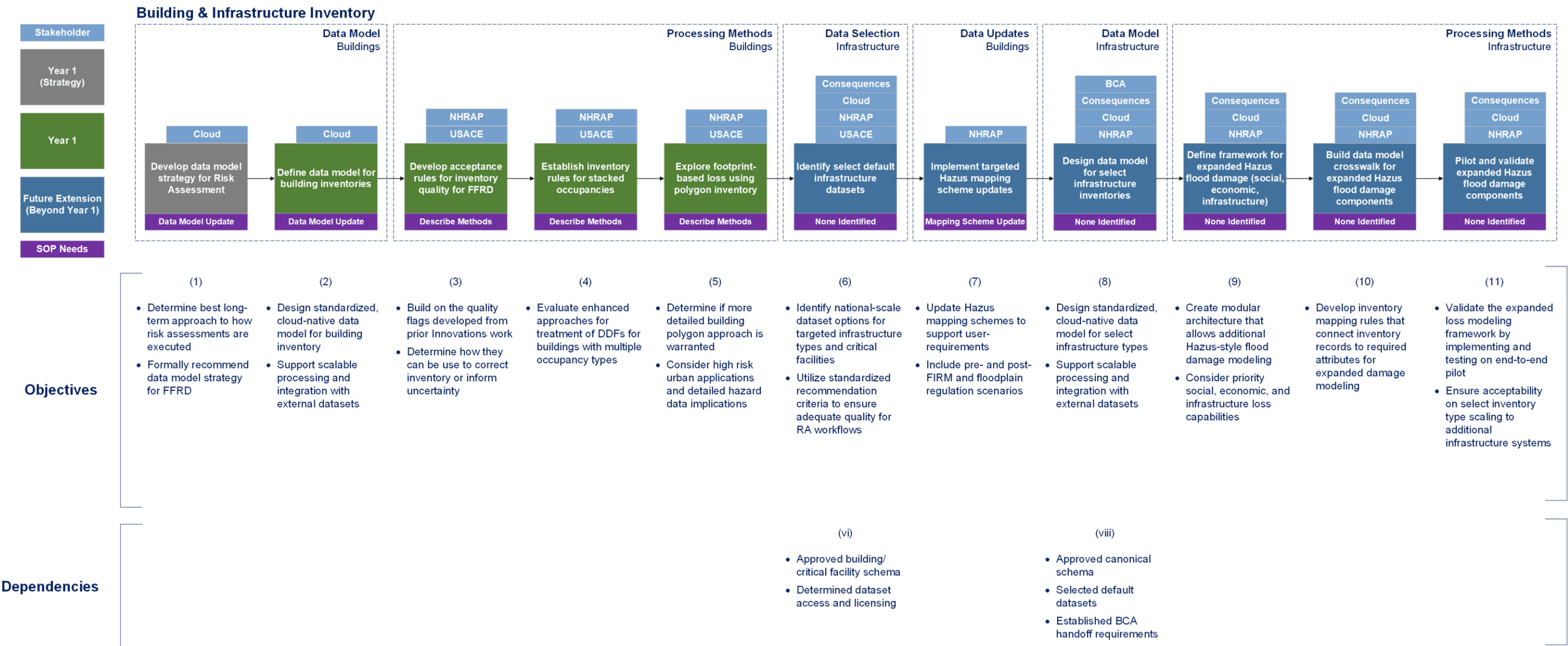
The Hazus program has a long history of defining and refining building and infrastructure inventory data for its locally-run, legacy software setup (**Table 5**). In recent years, this inventory refinement work has been focused on site-specific inventories aligned with the NSI and to determine methods to quantify any substantial sources uncertainty. Updates to-date have yet to address modular, cloud-native means of data referencing, including standardizing elements such as provenance and lineage, and methods investigations have yielded recommendations for minor method refinements before further widespread deployment.

Proposed Work

Additional data model and methods work is proposed to ensure inventory elements can be efficiently and reliably stored and processed under a cloud-native framework for FFRD (**Figure 10** and **Figure 11**). This includes ensuring that remaining edge cases for processing building attributes are finalized and that standards ensure inventories are modularly structured and can be readily versioned with further update. Resulting treatment will provide extendibility for third-party modification (e.g. by states and localities), 3D buildings, and fragility-based analysis, and allow maximum interoperability for a variety of potential loss tools and downstream users.

Table 5 – Building and Infrastructure Inventory: Completed Milestones

Milestone	Year	Documentation
Early Hazus Data Sources: Aggregated building data from Census and commercial sources are provided supporting losses at Census block and tract levels	Pre-2011	Hazus Archived Documentation
HIFLD Incorporation: Hazus’ partnership with the HIFLD provides nationwide, updated infrastructure layers, supporting site-specific analysis, specific infrastructure damage functions are limited	Pre-2011	Hazus Archived Documentation
Hazus 6.0: Migration from aggregated to site-specific building inventory (NSI). Includes full replacement costs and modifications to multi-family and building heights	2022	Hazus Archived Documentation
Hazus 6.1: Enhanced NSI-based dataset now includes fields representative of IBC era construction	2023	Hazus Archived Documentation
Hazus 7.0: Initial migration to ArcGIS Pro using aggregated enhanced NSI. Updates to foundation and height distributions based on observed data	2024	Hazus Documentation
Uncertainty Approaches: Prototype approaches to assess, quantify, and reduce building attribution uncertainties (e.g., foundation type and height), and establish criteria for identifying anomalies	2025	
NSI 2026 Release: Updated replacement cost methodology, integration of parcel data, improved alignment with building footprints, modernized file types and expanded to 134 million structures	2026	



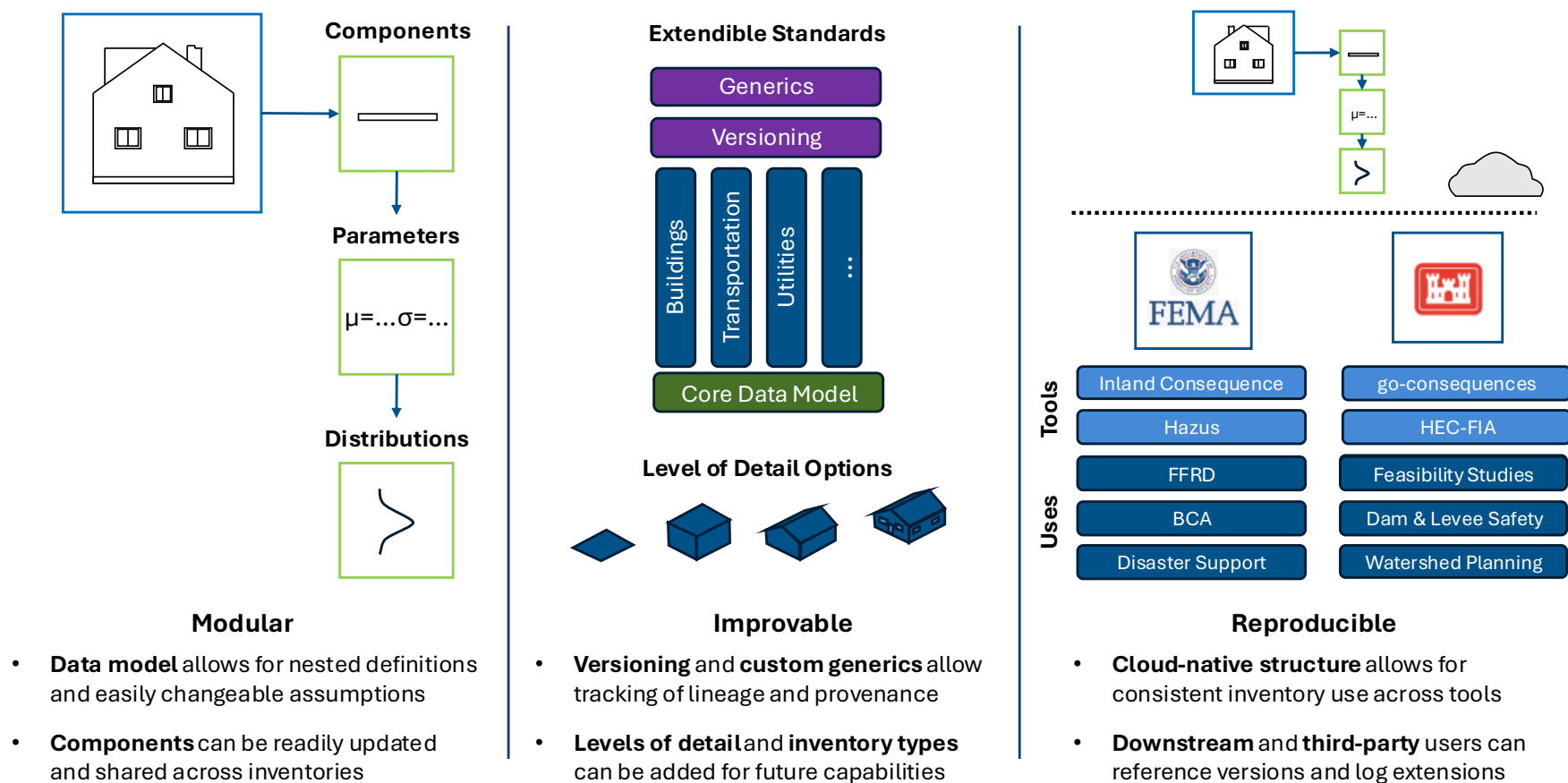


Figure 11 - Building and Infrastructure Inventory

Coastal Consequences

Work To-Date

Foundational research for coastal loss methods was completed as part of the development of the FFRD Coastal AAL Calculation Tool, originally developed in R (**Table 6**). This work was ported into Python to support integration with the Inland Consequences Tool and further refinement was put into coastal DDF selection, processing and interpolation methods, and strategies to incorporate future compound flooding computations. [Prior efforts](#) did not finalize or implement the processing and interpolation methods in any of the prototype tools and the framework for compound flooding has not moved beyond an initial set of recommendations.

Proposed Work

Additional data model and methods work is proposed to ensure prior refinements are prototyped and able to scale within operational tools and that compound flooding can be incorporated with greater detail in FFRD once further modeling decisions have been established (**Figure 12** and **Figure 13**). This includes preparing the data model for shared inland-coastal result outputs and refining interpolation methods, updating foundation class assignments. Such an approach will allow DDF libraries to be shared across coastal and inland risk assessments and utilized within a potential future tiered compound flooding screening methodology.

Table 6 - Coastal Consequences: Completed Milestones

Milestone	Year	Documentation
Coastal Loss R Tool: Create loss computation with wave and SWL uncertainty and node interpolation	2024	Coastal FFRD AAL Calculation Tool
Evaluate Coastal DDFs: Evaluate libraries of DDFs and assess DDFs by occupancy, wave height, etc.	2024	OpenHazard Supporting Research - Task 1
Recommend Coastal DDFs: Define mapping schemes for use by flood specific building types	2025	OpenHazard Supporting Research - Task 1
Barycentric Interpolation: Identify alternative interpolation method from coastal vector nodes	2025	Inland Consequence Tool - Task 4
Coastal Loss Python Tool: Port capabilities of R tool to the Inland Consequence repository in Python	2026	Inland Consequence Tool - Task 6
Tiered Compound Screening: Propose concept of tiered policy around levels of compound modeling detail	2026	OpenHazard Supporting Research - Task 4

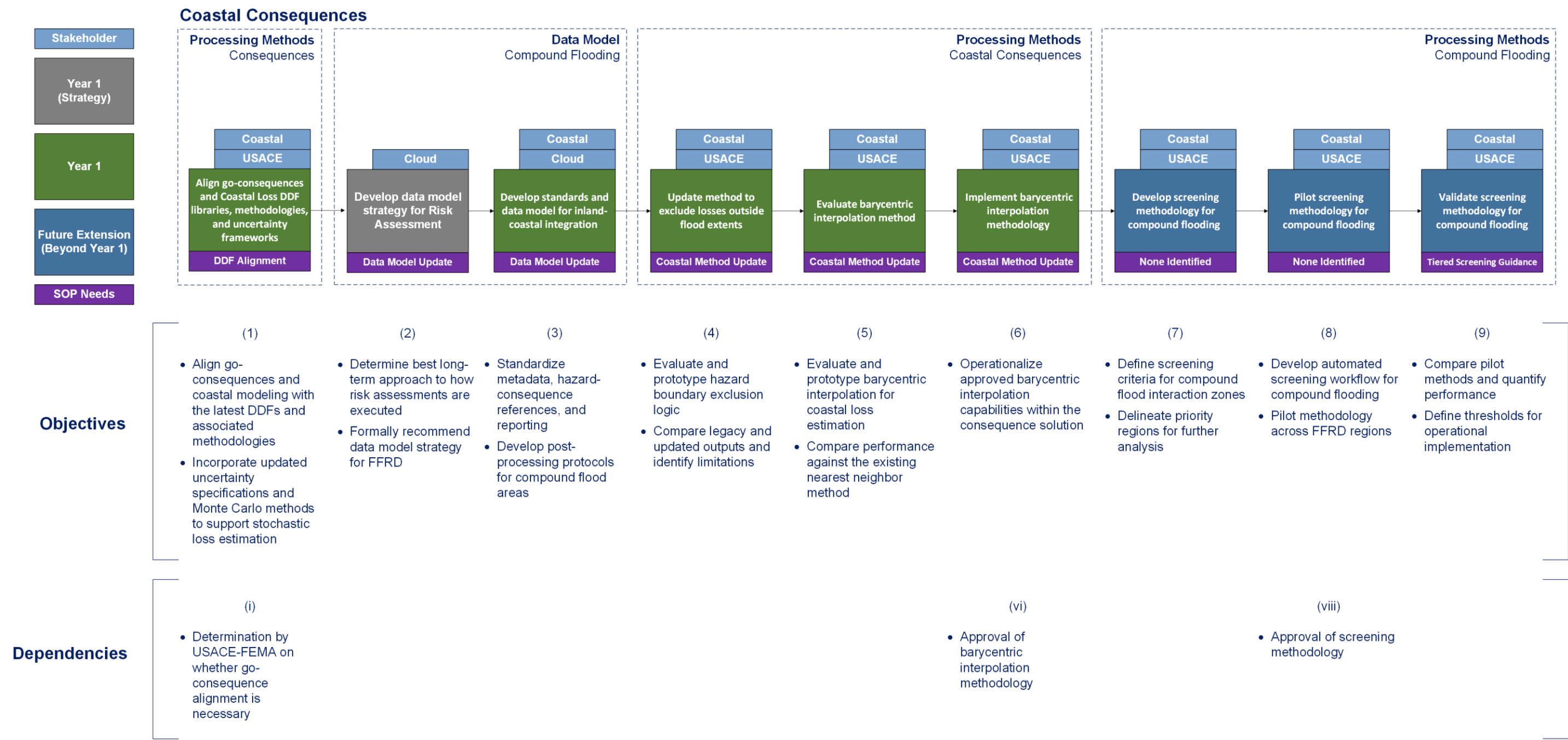


Figure 12 - Coastal Consequences Process Flowchart

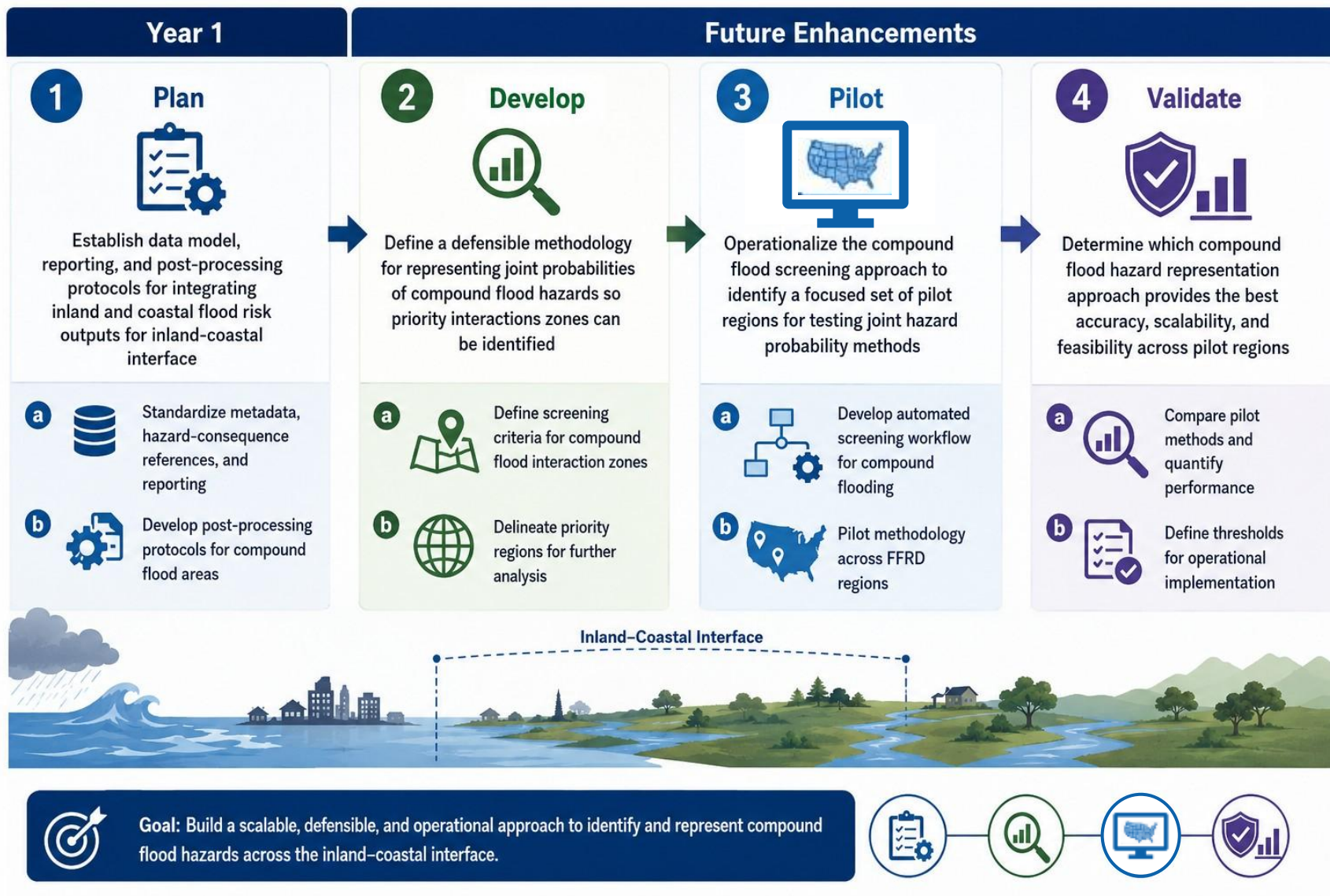


Figure 13 - Compound Flooding

Inland Consequences

Work To-Date

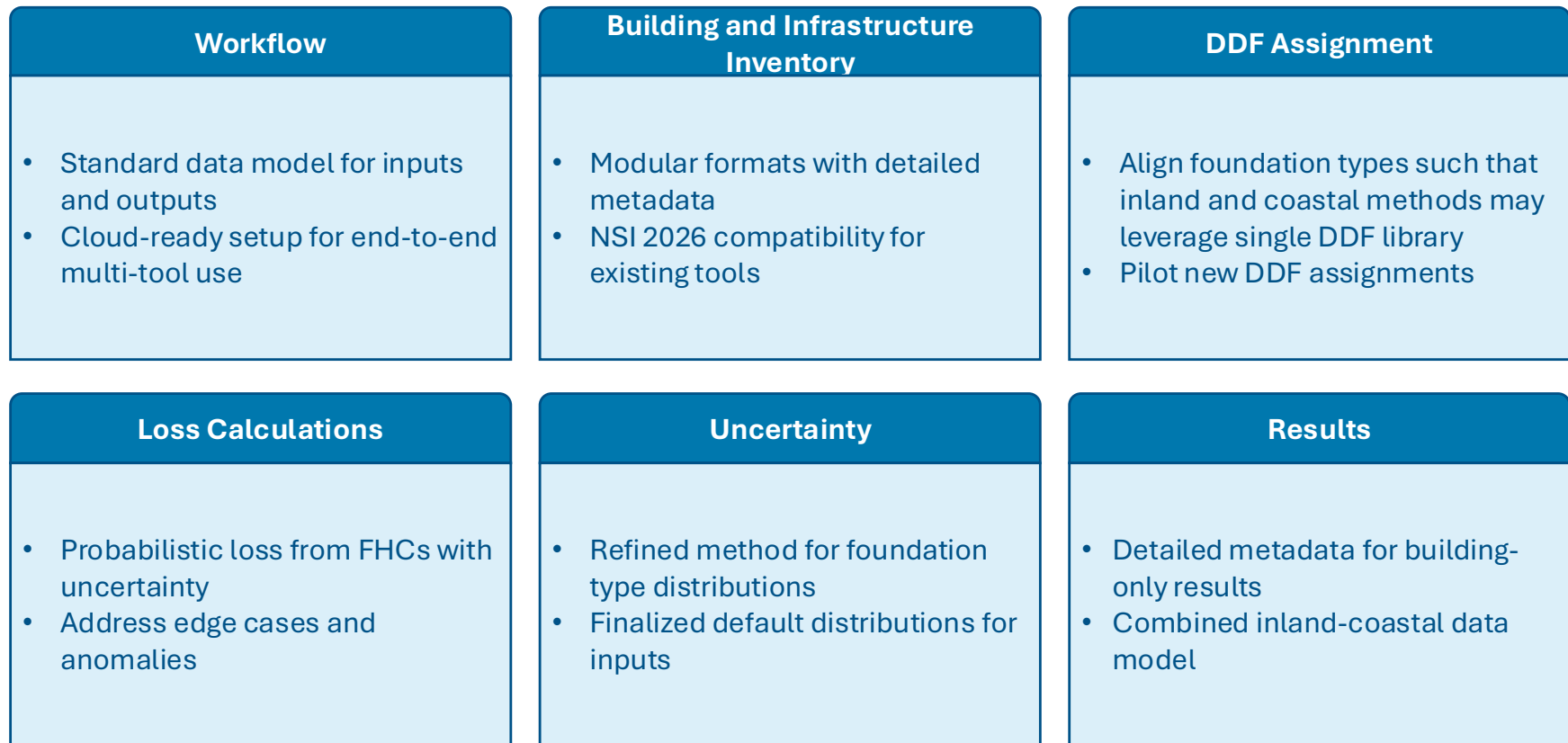
Inland Consequences was the focus of the last Innovations Standard Operation period, significant documentation and code are available on [GitHub](#) (**Table 7**). Inland consequence methodologies have undergone significant updates in recent years, primarily focused on validating model results against NFIP claims data and improving damage function assignments and uncertainty characterization. Most recently, updates were made to the DDF Library and damage function assignment methodology in response to reviewer feedback, including a revised approach for assigning contents damage functions.

Proposed Work

There are no proposed tasks related solely to inland consequences, however, many of the tasks in other workstreams will impact inland flooding hazard identification and risk assessment (**Figure 14**).

Table 7 - Inland Consequences: Completed Milestones

Milestone	Year	Documentation
Expanded Single Family Inland Flood DDFs: Expand FEMA PFRA DDF approaches from coastal to single family inland flooding	2021	2021 Inland Single Family Residential DDF Report
DDF Validation: Initial investigation of DDF validation from NFIP Claims Data	2024	2024 NFIP Claims Data Exploration DDFs Report
DDF Library & Damage Function Assignment: Initial development of comprehensive DDF Library and propose new damage function assignment using flood-specific building types and revised foundation types	2024	OpenHazarus Supporting Research Report – Task 1 Sec. 3
Evaluate and Recommend DDFs for OpenHazarus/FFRD: Evaluate 2024 DDF Library and proposed new damage function assignment for feasibility for application to OpenHazarus and FFRD	2025	2025 DDF Validation Report
DDF Validation: Additional investigation potentially updating DDFs from NFIP Claims Data	2025	OpenHazarus Supporting Research - Task 1
Pilot test inland flood for FFRD: Test 2024 DDF damage function assignment approach for structure damages for inland flood FFRD prototype for Duwamish, including testing uncertainty approaches	2025	OpenHazarus Supporting Research - Tasks 1, 2, and 3
DDF Library and Damage Function Assignment: Updated in Spring 2026 comprehensive DDF Library and damage function assignment approach based on reviewer feedback, including revised contents damage function assignment approach	2026	Inland Consequence Tool Report - Task 5 Sec. 8

**Figure 14 - Impacts to Inland Consequences Methodology**

Downstream Uses

Work To-Date

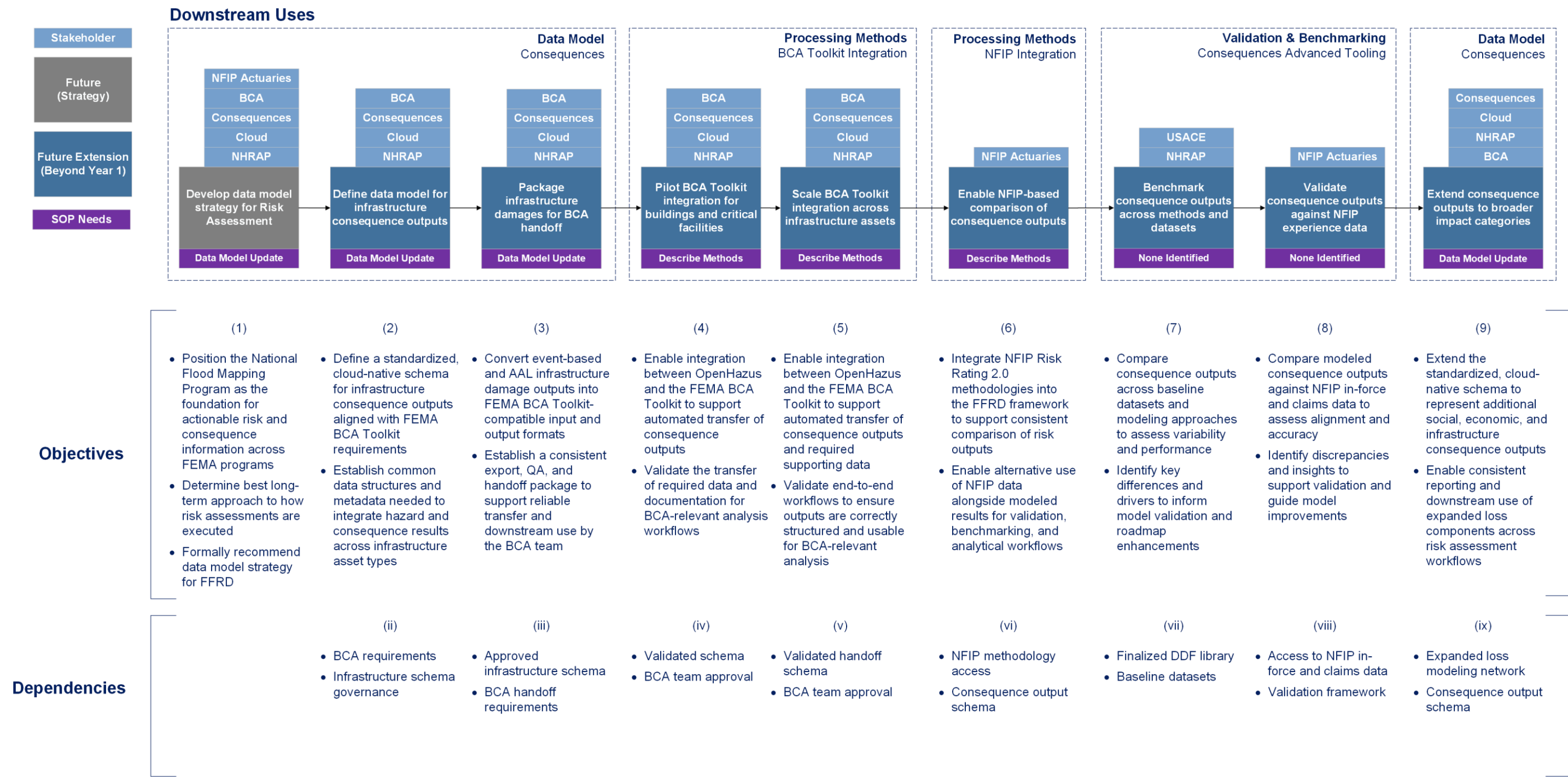
To date, Downstream Uses work has focused on understanding how FEMA can use consequence model results in practical workflows, including BCA, NFIP actuarial analysis, mitigation planning, and reporting (**Table 8**). Earlier work looked at how to estimate mitigation costs and return on investment, and how those results could support FEMA's Mitigation Project Designer process. Other research reviewed damage methods, data needs, and how FFRD information could be useful for NFIP and BCA users. The Inland Consequences work builds on this by gathering feedback from stakeholders, documenting BCA and NFIP needs, reviewing current tool limitations, evaluating existing consequences tools, and developing a Python-based prototype approach for calculating AALs. The draft report notes that BCA users need defensible structure-level AALs, return-period damages, clear calculation logic, uncertainty information, metadata-rich exports, and outputs that can work with FEMA BCA tools. NFIP users need transparent assumptions, losses broken out by peril and coverage type, geographic rollups, and structure-level AALs that can support actuarial analysis.

Proposed Work

Future work will focus on expanding the value of the National Flood Mapping Program (NFMP) by transforming flood hazard information into standardized consequence products that support FEMA and SLTT decision-making (**Figure 15**). This includes developing consistent outputs for BCA and NFIP applications, enabling integration with the FEMA BCA Toolkit, and supporting the use of AAL and event-based damage estimates (**Figure 16**). Future efforts will also evaluate how Risk Rating 2.0 methodologies, policy, and claims data can be used to benchmark and validate modeled losses. Collectively, this work will strengthen NFMP's role as a foundational source of flood risk information, supporting actuarial analysis, mitigation investment decisions, and reporting on flood-related economic, social, and infrastructure impacts.

Table 8 - Downstream Uses: Completed Milestones

Milestone	Year	Documentation
Mitigation Methods: Create methods to cost and compute ROI for common mitigation activities	2023	FEMA Mitigation Project Designer Loop
Mitigation Compute: Recommend compute framework for consequence mitigation wrapper	2024	FEMA Mitigation Project Designer Loop
Fragility-based DDF Review: Assess the state of component-level damage methods and input needs	2024	OpenHazard Supporting Research - Task 2
FFRD-NFIP Engagement: Gather stakeholder feedback on FFRD data use cases and application needs	2024	FFRD Engagement Recommend. Summary
FFRD-BCA Engagement: Explore the use of FFRD data within the BCA Toolkit and identify future needs	2025	FFRD BCA Final Report
Consequence Tool Engagement: Gather stakeholder feedback on current tool limitations and requirements	2025	Inland Consequences Tool – Task 1
Consequence Tool Review: Evaluate existing tool capabilities and pros and cons for prototype integration	2025	Inland Consequences Tool – Task 2
Standardized Python Approach: Develop prototype Inland Consequence tool for AAL methods	2025	Inland Consequences Tool – Task 3
Scalable Compute Needs: Identify data standard and cloud compute needs to scale methods	2025	OpenHazard Supporting Research – Task 4



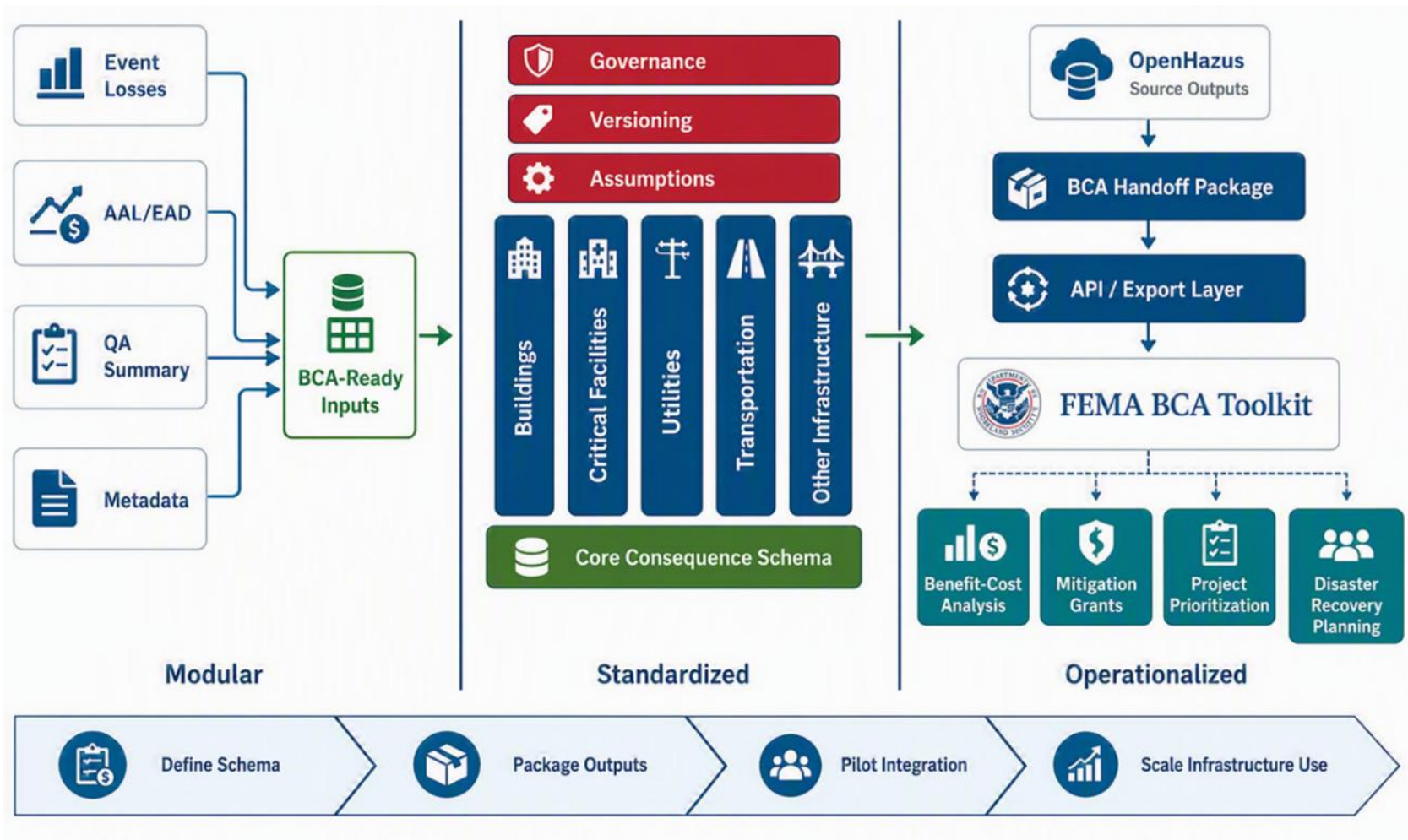


Figure 16 - BCA Toolkit Integration

Recommended Immediate Start Activities

- **Task: Develop data model strategy for Risk Assessment**
 - **Objectives:**
 - Determine best long-term approach to how risk assessments are executed
 - Formally recommend data model strategy for FFRD
 - **Stakeholders:**
 - Cloud
 - **SOP Needs:**
 - Data model update

- **Task: Refine methods to estimate regional distribution for foundation types**
 - **Objectives:**
 - Standardize method for estimating regional foundation type parameters.
 - Account for uncertainty and geographic sparsity considerations
 - **Stakeholders:**
 - USACE
 - **SOP Needs:**
 - Default assumptions

- **Task: Run frequency-based AAL for Trinity Watershed to use as baseline for validation and comparison of results**
 - **Objectives:**
 - Develop baseline consequence results for Trinity using the latest modeling methods
 - Evaluate the sensitivity of results to alternative building inventories.
 - Validate modeled losses against NFIP claims and in-force policy data to refine methodologies
 - **Stakeholders:**
 - Cloud
 - **SOP Needs:**
 - None identified